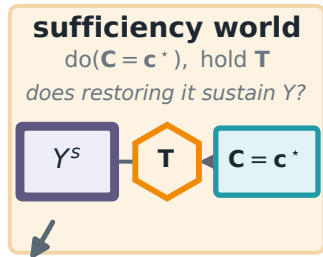
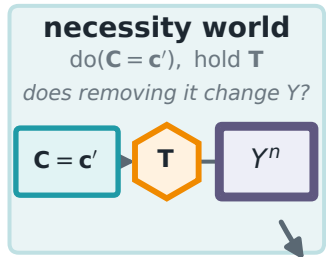


(a) Necessity and sufficiency on a shared draw

one shared draw $\mathbf{u} \sim P_{\mathbf{u}}$



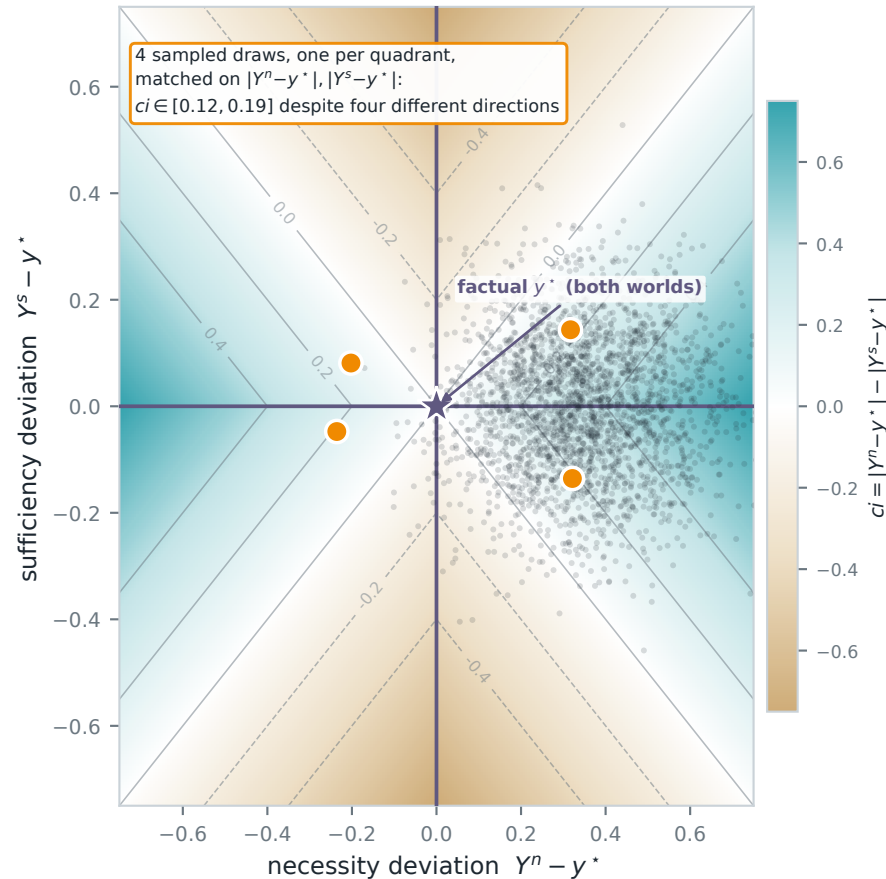
$$ci(Y^s, Y^n, y^*)$$

$$\mathbb{E}_{\Gamma \otimes \Delta \otimes P_{\mathbf{u}}}[ci] = \text{PCI score for } X_k$$

OBCB instance

X_k = gender, $\mathbf{T} = \{\text{check-failed}\}$ held at factual,
 $\mathbf{c}^*$ = female (restore), $\mathbf{c}'$ = male (alternative)

(b) Same score, four quadrants



(c) Summary: score = mean of sampled draws

